gpeyt1-28 · Unconfined well (Thiem)

GEO PE · Long Calculation Problems I — Topic 6: Groundwater & Seepage

Problem

A fully penetrating dewatering well in an unconfined sand-and-gravel aquifer (K = 45 ft/day) resting on horizontal impermeable clay is pumped to steady state. Observation wells at r1 = 25 ft and r2 = 150 ft show saturated thicknesses (water level above the clay) of h1 = 38.0 ft and h2 = 41.5 ft. The well radius is 0.5 ft, the static saturated thickness was 44 ft, and the pump is set 5 ft above the clay.

Most nearly, the pumping rate is:

A. 50,551 ft³/day
B. 21,954 ft³/day
C. 967 ft³/day
D. 43,908 ft³/day

Given equations / data

> Unconfined (Dupuit–Thiem): $Q = \pi K(h2^2 - h1^2)/\ln(r2/r1)$

Worked solution

1. $h2^2 - h1^2 = 1722.25 - 1444.00 = 278.25\ ft^2$
2. $\ln(150/25) = 1.7918$
3. $Q = \pi \times 45 \times 278.25/1.7918 = 21{,}954\ ft^3/day$ (≈ 114 gpm)

Final answer

21,954 ft³/day

Why the other values are wrong

- **50,551 ft³/day** — used log10
- **967 ft³/day** — squared the difference instead of differencing squares
- **43,908 ft³/day** — used 2π (confined form)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §6.6.4 Well Analysis–Steady State (unconfined aquifer); §3.16.5 Construction Dewatering

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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